

Critical Room Tryhackme Writeup

This is a writeup for the tryhackme room which focuses on Digital Forensics and Incident Response. In this room a memory dump (which is an image of random access memory) is analyzed to solve a case where an important document has been encrypted.

FTK Imager, a tool that helps with imaging and forensics will be used for this task.



Once logged into the analyst account, the home directory showed that it had two files. The memory dump was named memdump.mem. In addition, there is volatility 3 a volatile memory analysis tool.

```
analyst@ip-10-10-135-148:~$ pwd
/home/analyst
analyst@ip-10-10-135-148:~$ ls
memdump.mem  volatility3-2.5.2
analyst@ip-10-10-135-148:~$
```

The first three questions were based on the content of the lesson and answered according to that.

1. Which plugin can help us to get information about the OS running on the target machine?
[answer found in reading content]
2. Which tool referenced above can help us take a memory dump on a Linux OS?
[answer found in reading content]
3. Which command will display the help menu using Volatility on the target machine?
[hint the command to call volatility and the help argument]

The next step was to get more information on the file we were analyzing. Since it was a windows os, it was possible to use the [windows.info](#) plugin by typing in the following command; vol -f memdump.mem windows.info

```
analyst@ip-10-10-135-148:~$ vol -f memdump.mem windows.info
Volatility 3 Framework 2.5.2
Progress: 100.00 PDB scanning finished
Variable Value
Kernel Base 0xf8066161b000
DTB 0x1ad000
Symbols file:///home/analyst/volatility3-2.5.2/volatility3/symbols/windows/ntkrnlmp.pdb/4DBE144182FF4156845CD3BD8B654E56-1.json.xz
Is64Bit True
IsPAE False
layer_name 0 WindowsIntel32e
memory_layer 1 FileLayer
KdVersionBlock 0xf8066222a400
Major/Minor 15.19041
MachineType 34404
KeNumberProcessors 2
SystemTime 2024-02-24 22:52:52
NtSystemRoot C:\Windows
NtProductType NtProductWinNt
NtMajorVersion 10
NtMinorVersion 0
PE MajorOperatingSystemVersion 10
PE MinorOperatingSystemVersion 0
PE Machine 34404
PE TimeDateStamp Sat Jan 13 03:45:32 2085
```

All of this info was thus used to answer the following questions:

4. Is the architecture of the machine x64 (64bit) Y/N?
5. What is the Version of the Windows OS
6. What is the base address of the kernel?

Next it was time to analyse some of the interesting things in the memory dump. Thus the windows.netsat plugin was used to analyze the network activity.

```
analyst@ip-10-10-135-148:~$ vol -f memdump.mem windows.netsat
Volatility 3 Framework 2.5.2
Progress: 100.00 PDB scanning finished
Offset Proto LocalAddr LocalPort ForeignAddr ForeignPort State PID Owner Created
0xe50ed9170ac0 TCPv4 192.168.182.139 49723 192.16.49.85 80 CLOSE_WAIT 4780 SearchApp.exe 2024-02-24 22:48:49.000000
0xe50ed8a4ca20 TCPv4 192.168.182.139 49814 52.142.223.178 443 SYN_SENT 368 svchost.exe 2024-02-24 22:52:43.000000
0xe50ed9275a20 TCPv4 192.168.182.139 3389 192.168.182.150 49253 ESTABLISHED 744 svchost.exe 2024-02-24 22:47:52.000000
0xe50ed9df3a20 TCPv4 192.168.182.139 49745 13.107.213.254 443 CLOSE_WAIT 4780 SearchApp.exe 2024-02-24 22:50:42.000000
0xe50ed8c52a20 TCPv4 192.168.182.139 49719 23.222.237.202 443 CLOSE_WAIT 4780 SearchApp.exe 2024-02-24 22:48:47.000000
0xe50ed9427a20 TCPv4 192.168.182.139 49694 20.7.1.246 443 ESTABLISHED 368 svchost.exe 2024-02-24 22:47:54.000000
0xe50ed83ea4d0 TCPv4 192.168.182.139 49743 23.222.237.203 443 CLOSE_WAIT 4780 SearchApp.exe 2024-02-24 22:50:39.000000
0xe50edac57a20 TCPv4 192.168.182.139 49712 152.199.55.200 443 CLOSE_WAIT 4780 SearchApp.exe 2024-02-24 22:48:06.000000
0xe50ed9508a20 TCPv4 192.168.182.139 49744 23.222.237.203 443 CLOSE_WAIT 4780 SearchApp.exe 2024-02-24 22:50:39.000000
0xe50ed6390cb0 TCPv4 0.0.0.0 135 0.0.0.0 0 LISTENING 896 svchost.exe 2024-02-24 22:47:36.000000
0xe50ed6390cb0 TCPv6 :: 135 :: 0 LISTENING 896 svchost.exe 2024-02-24 22:47:36.000000
0xe50ed6391910 TCPv4 0.0.0.0 135 0.0.0.0 0 LISTENING 896 svchost.exe 2024-02-24 22:47:36.000000
0xe50ed6391bd0 TCPv4 192.168.182.139 139 0.0.0.0 0 LISTENING 4 System 2024-02-24 22:47:36.000000
0xe50ed818d310 TCPv4 0.0.0.0 445 0.0.0.0 0 LISTENING 4 System 2024-02-24 22:47:37.000000
0xe50ed818d310 TCPv6 :: 445 :: 0 LISTENING 4 System 2024-02-24 22:47:37.000000
0xe50ed36a59f0 TCPv4 0.0.0.0 3389 0.0.0.0 0 LISTENING 744 svchost.exe 2024-02-24 22:47:36.000000
0xe50ed36a59f0 TCPv6 :: 3389 :: 0 LISTENING 744 svchost.exe 2024-02-24 22:47:36.000000
0xe50ed36a5730 TCPv4 0.0.0.0 3389 0.0.0.0 0 LISTENING 744 svchost.exe 2024-02-24 22:47:36.000000
0xe50ed86f80d0 TCPv4 0.0.0.0 5040 0.0.0.0 0 LISTENING 1220 svchost.exe 2024-02-24 22:47:39.000000
0xe50ed818e0d0 TCPv4 0.0.0.0 7680 0.0.0.0 0 LISTENING 5572 svchost.exe 2024-02-24 22:47:44.000000
0xe50ed818e0d0 TCPv6 :: 7680 :: 0 LISTENING 5572 svchost.exe 2024-02-24 22:47:44.000000
0xe50ed6390310 TCPv4 0.0.0.0 49664 0.0.0.0 0 LISTENING 664 lsass.exe 2024-02-24 22:47:36.000000
0xe50ed6390310 TCPv6 :: 49664 :: 0 LISTENING 664 lsass.exe 2024-02-24 22:47:36.000000
```

It was noted that there was a connection established to port 3389 by a foreign ip address 192.168.182.158 which could be a matter of concern.

Next, processes were analyzed using the windows.pstree plugin and the command vol -f memdump.mem windows.pstree.

```
analyst@ip-10-10-135-148:~$ vol -f memdump.mem windows.pstree
```

Volatility 3 Framework 2.5.2

Progress: 100.00 PDB scanning finished

PID	PPID	ImageFileName	Offset(V)	Threads	Handles	SessionId	Wow64	CreateTime	ExitTime
4	0	System	0xe50ed3687040	150	-	N/A	False	2024-02-24 22:47:35.000000	N/A
* 312	4	smss.exe	0xe50ed68b0040	2	-	N/A	False	2024-02-24 22:47:35.000000	N/A
* 1600	4	MemCompression	0xe50ed379e280	50	-	N/A	False	2024-02-24 22:47:36.000000	N/A
* 92	4	Registry	0xe50ed36ed080	4	-	N/A	False	2024-02-24 22:47:31.000000	N/A
424	400	csrss.exe	0xe50ed67d7140	9	-	0	False	2024-02-24 22:47:35.000000	N/A
500	400	wininit.exe	0xe50ed7366080	2	-	0	False	2024-02-24 22:47:35.000000	N/A
* 664	500	lsass.exe	0xe50ed7360080	8	-	0	False	2024-02-24 22:47:35.000000	N/A
* 776	500	fontdrvhost.ex	0xe50ed7c69140	6	-	0	False	2024-02-24 22:47:35.000000	N/A
* 636	500	services.exe	0xe50ed73d3080	6	-	0	False	2024-02-24 22:47:35.000000	N/A
** 896	636	svchost.exe	0xe50ed7d112c0	9	-	0	False	2024-02-24 22:47:36.000000	N/A
** 1924	636	svchost.exe	0xe50ed73ab2c0	5	-	0	False	2024-02-24 22:47:36.000000	N/A
** 3464	636	svchost.exe	0xe50ed88e3080	7	-	1	False	2024-02-24 22:47:39.000000	N/A
** 7312	636	SecurityHealth	0xe50ed9af1280	10	-	0	False	2024-02-24 22:47:56.000000	N/A
** 2964	636	dllhost.exe	0xe50ed858d280	14	-	0	False	2024-02-24 22:47:37.000000	N/A
** 3348	636	svchost.exe	0xe50ed8b722c0	6	-	0	False	2024-02-24 22:47:39.000000	N/A
** 7060	636	WUDFHost.exe	0xe50ed9ad41c0	9	-	0	False	2024-02-24 22:47:53.000000	N/A
** 792	636	svchost.exe	0xe50ed7c85240	13	-	0	False	2024-02-24 22:47:35.000000	N/A
*** 3336	792	LockApp.exe	0xe50ed99b70c0	19	-	1	False	2024-02-24 22:47:46.000000	N/A
*** 5008	792	RuntimeBroker.	0xe50ed916c300	12	-	1	False	2024-02-24 22:47:40.000000	N/A
*** 4368	792	RuntimeBroker.	0xe50edab30080	5	-	1	False	2024-02-24 22:49:46.000000	N/A
*** 4500	792	RuntimeBroker.	0xe50ed883f0c0	4	-	1	False	2024-02-24 22:47:40.000000	N/A
*** 6292	792	PhoneExperienc	0xe50ed99da080	17	-	1	False	2024-02-24 22:47:47.000000	N/A
*** 7188	792	smartscreen.ex	0xe50ed9e29300	7	-	1	False	2024-02-24 22:47:56.000000	N/A
*** 7452	792	WmiPrvSE.exe	0xe50eda204280	6	-	0	False	2024-02-24 22:47:57.000000	N/A
*** 5920	792	ShellExperienc	0xe50ed96c71c0	12	-	1	False	2024-02-24 22:47:45.000000	N/A
*** 2848	792	FileCoAuth.exe	0xe50ed95d2080	6	-	1	False	2024-02-24 22:52:05.000000	N/A
*** 4780	792	SearchApp.exe	0xe50ed9299080	47	-	1	False	2024-02-24 22:47:40.000000	N/A
*** 2864	792	dllhost.exe	0xe50ed855c2c0	6	-	0	False	2024-02-24 22:47:37.000000	N/A

The lesson content provided a valuable piece of information; a list of the most common processes that run on a windows system. Any deviation from this baseline would be regarded as potentially malicious activity. It was noted that there was a process named: critical-updat which is the parent process of updatер.exe which is not part of windows processes.

**** 3384	7960	conhost.exe	0xe50edab37080	4	-	1	False
**** 1648	7960	critical_updat	0xe50ed94c1080	5	-	1	False
*****	1648	updatер.exe	0xe50edab53080	6	-	1	False
*** 6460	3196	FTK Imager.exe	0xe50edad09080	19	-	1	False

(Image taken from the TryHackMe critical room)

The next questions were answered as follows:

- Using the plugin "windows.netscan". Can you identify the destination IP address where a connection is established on port 80?
Command: vol -f memdump.mem windows.netscan then looking through the connections for an established connection to port 80.
- Using the plugin "windows.netscan," can you identify the program (owner) used to access through port 80?
[answer is in the results of the above command]
- Analyzing the processes present on the dump, what is the PID of the child process of critical_updat?
[answer can be found using the windows.pstree plugin]
- What is the time stamp time for the process with the truncated name critical_updat?
[looking through the results of the above action led to the answer]

11. Analyzing the "windows.filescan" output, what is the full path and name for critical_updat?

The next action was to find some interesting data by looking at the files accessed in the memory dump. The plugin windows.filescan was used for this activity and the results were outputted to a file. The full command was `vol -f memdump.mem windows.filescan > filescan_results`. Grep was also used to filter through the results by utilizing the command:

```
analyst@ip-10-10-107-191:~$ vol -f memdump.mem windows.filescan > filescan_results
analyst@ip-10-10-107-191:~$ ^C PDB scanning finished
```

Cat filescan_results | grep "critical_updat"

```
analyst@ip-10-10-107-191:~$ cat filescan_results | grep "critical_updat"
0xe50edaa3ac20 \Users\user01\Documents\critical_update.exe 216
```

12. Analyzing the "windows.mftscan.MFTScan" what is the Timestamp for the created date of important_document.pdf?

To get the answer, an mftscan was conducted using the command: `vol -f memdump.mem windows.mftscan.MFTScan > mftscan_results`

```
analyst@ip-10-10-107-191:~$ vol -f memdump.mem windows.mftscan.MFTScan > mftscan_results
Progress: 100.00 PDB scanning finished
```

Next it was time to filter for the result I wanted using `cat mftscan_results | grep "important_document"`

```
analyst@ip-10-10-107-191:~$ cat mftscan_results | grep "important_document"
* 0xd389c5fbad28 FILE 111083 2 File Archive FILE_NAME 2024-02-24 20:39:42.000000 2024-02-24 20:39:42.000000 2024-02-24 20:39:42.00
0000 2024-02-24 20:39:42.000000 important_document.pdf
* 0xd389c664ad28 FILE 110635 2 File Archive FILE_NAME 2024-02-24 17:11:21.000000 2024-02-24 22:52:07.000000 2024-02-24 22:52:07.00
0000 2024-02-24 22:52:07.000000 important_document.lnk
* 0xd389db802930 FILE 110635 2 File Archive FILE_NAME 2024-02-24 17:11:21.000000 2024-02-24 22:52:07.000000 2024-02-24 22:52:07.00
0000 2024-02-24 22:52:07.000000 important_document.lnk
```

From this I was able to get my answer for the question.

13. Analyzing the updater.exe memory output, can you observe the HTTP request and determine the server used by the attacker?

The following actions were taken to answer this question.

Firstly, I used the command `vol -f memdump.mem -o . windows.memmap --dump --pid 1612`

Then I filtered through it using grep:

```
strings pid.1612.dmp |grep -B 10 -A 10
"http://key.critical-update.com/encKEY.txt"
```

The http request thus contained information about the server used.

```
<li><a href= critical_update.exe">critical_update.exe</a></li>
<li><a href="dumps/">dumps/</a></li>
<li><a href="encKEY.txt">encKEY.txt</a></li>
<li><a href="updater.exe">updater.exe</a></li>
</ul>
<hr>
</body>
</html>
@s1/0/_dk http://critical-update.com http://critical-update.com http://key.critical-update.com/encKEY.txt
HTTP/1.0 200 OK
Server: SimpleHTTP/0.6 Python/3.10.4
Date: Sat, 24 Feb 2024 22:52:40 GMT
Content-type: text/plain
Content-length: 9
Last-Modified: Fri, 23 Feb 2024 22:56:51 GMT
192.168.182.128
cafebabe
ul1/0/_dk https://microsoft.com https://microsoft.com https://edge.microsoft.com/entityextractiontemplates/api/v1/assets/find-assets?name=product_category_en&version=
1.*.*&channel=stable&key=d414dd4f9db345fa8003e32adc81b362
1/0/_dk https://critical-update.com https://critical-update.com https://key.critical-update.com/encKEY.txt/
```

And thus I completed the exercise.

Lessons Learned

- The criticality of Random access memory and how important it is to get a memory dump for forensics.
- The use of Volatility 3 tool to analyze a memory as well as interesting plugins that can aid in investigation.
- How to get Os information, process trees, network activity etc from a memory dump
- The practicality of knowing what to look for so you can use grep to filter otherwise extensive results.